



A woman looking at a wearable fitness device, while out on a run

sports content on social media platforms such as Facebook and Twitter; again, all within the past one year.

The figures also suggest that sports is a major driver for buying bigger televisions and 62 per cent of self-described sports fans own a television with a screen size of 101.6cm (40-inches) or more as compared to only 38 per cent in the general population.

Electronic billboards are used in stadiums and arenas by sports teams to resell advertising spaces to their sponsors and advertisers. These are specifically developed for use at stadiums as classic electronic billboards. The modular system allows assembling in any shape and size. With the use of electronics based recording and displaying of ongoing sports activities, their outcome in the stadium for online viewers has reached remarkable heights of satisfaction.

Wearable electronics

Training, especially in sports, takes discipline. But if we arm ourselves with the right fitness wearable, we will be more invigorated to stay on course with our programme and see the results we have been striving to achieve.

Electronics based technology is revolutionising sports and training.

Materials including fibres, films and coatings that react to electrical, optical or magnetic signals provide embedded intelligence to knit woven or non-woven textile structures. Integration of electronics with textiles brings a new frontier in functionality to one of the most common materials in our daily lives.

Companies are looking to seamlessly integrate electronics with textile structures, changing the way people benefit from technology in sports and develop wearable electronics that can be used to enhance sports performance. The goal is to change the way people interact with technology, and develop and commercialise comfortable, practical, scalable and affordable sports solutions.

Wearables can take on a number of different forms.

A tough fabric wearable sensor is rugged, accurate and slim enough to be invisibly integrated in the wearer's clothing or on the body. The innovative strain-locking mechanism allows the sensor to not get stretched beyond its maximum stretch range. As a result, those wearing these can move in any way they wish and take part in rough activities including vigorous sports.

Athletic garments and gear are critical to the success of athletes in

sports ranging from swimming to hockey. Engineers design these materials to maximise athletes' performance and keep them safe. In a sport like alpine skiing, where seconds can make the difference between which team takes home the gold, uniforms can have a huge impact.

GPS analysis of trial runs on the course has helped guide the placement of seams and zippers to minimise drag or resistance. The super-sleek suits even underwent testing in a wind tunnel. Padding within the suits was also made from a specialised material designed to harden on impact, to better protect wearers.

Helmets are a critical piece of sports equipment for preventing traumatic brain injuries in high-impact sports like football and hockey. Engineers have developed smarthelmets to further prevent and reduce concussions. These helmets contain sensors that measure the force and location of the impact and send alerts about possible concussions via a smartphone app to coaches on the sideline.

Engineers are even considering adding strong magnets inside smart-helmets, which would help disperse the energy of impact using their repellant properties. Smarthelmets have the potential to reduce the severity of injuries and improve the time in which players receive medical attention.

A new type of wearable technology that provides information about a player's jumping performance, known as VERT, has been developed. VERT uses a small sensor-equipped device, which is worn on the waistband of a player's shorts.

Using a computer processor and a special algorithm, or set of rules, VERT can calculate jumping data such as the number of jumps, height of jumps and average and highest jumps. Data is sent via Bluetooth to an app on the coach's mobile device. The app provides information on not only the player's jumping performance but also on whether he or she is at risk for possible injury due to fatigue.